



Valuation and Forecasting Model for Residential Properties

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Capstone Presentation

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Agenda

- The Disconnect in Spatiotemporal Valuation
- Component I: Micro-Valuation (LME)
- Component II: Macro-Forecasting
- Robust Evaluation & Key Results
- Business Impact & Blueprint

The current real estate valuation systems are fragmented, failing to simultaneously integrate Macro-level market momentum with Micro-level local property desirability. This gap leads to inaccurate pricing and undefined risk ranges for institutional investors.

A Two Component Framework

Market Engine:

- 1. Predicts the future market trajectory and risk**
- 2. Addresses market fragmentation and lacks of risk quantification.**

Property Engine :

- 1. Calculates individual property's intrinsic and latent value.**
- 2. Addresses inadequate neighborhood adjustments and spatial fragmentation.**

Market Engine

Probabilistic ZIP-level price forecasts (1-2 months ahead)

We build a ZIP×month price index and train a multi-task tabular neural network that forecasts the **distribution** of future price moves, not just a single point estimate.

- **Log-return targets (what we forecast):**

$$R_{z,t}(h) = \log(1 + P_{z,t}(h)) - \log(1 + P_{z,t}), h \in \{1, 2\}$$

Where: $P_{z,t}(1) = P_{z,t+1}$ and $P_{z,t}(2) = P_{z,t+2}$

- **Model mapping (what the network does):**

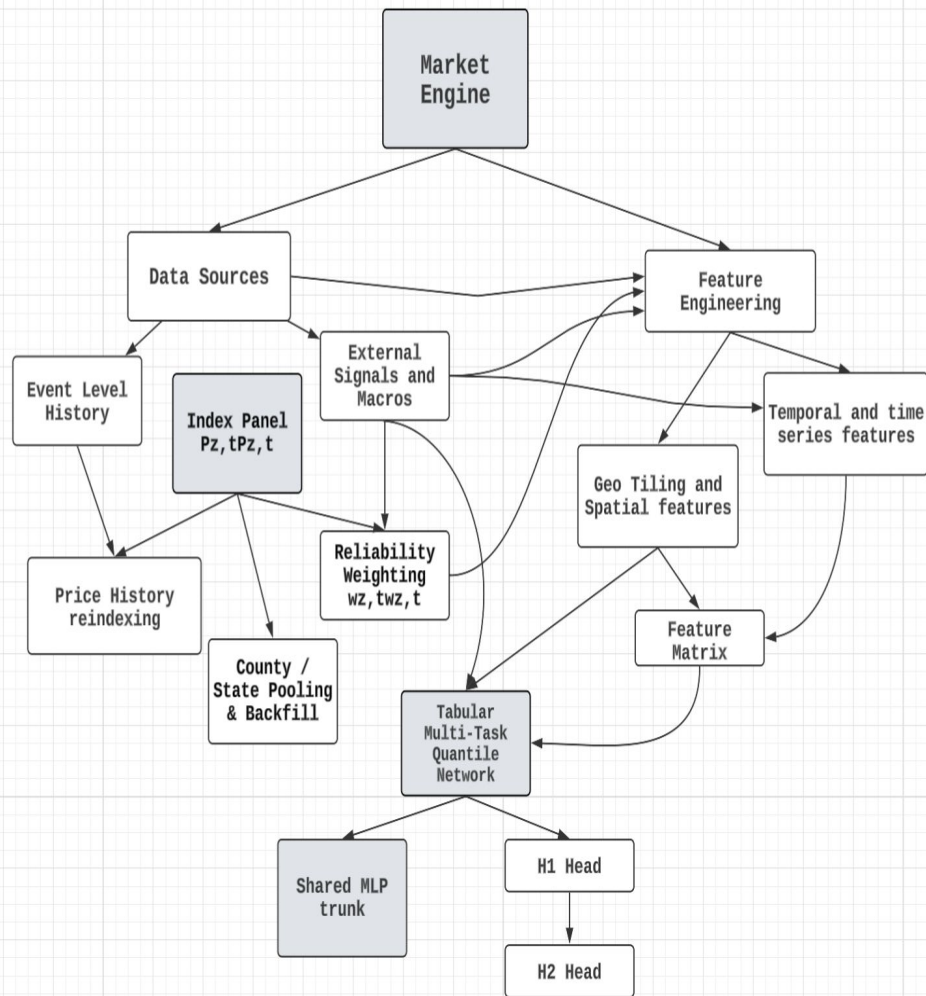
$$f_{\theta}(xz, t) \rightarrow \{Q^{\tau}(1)(z, t), Q^{\tau}(2)(z, t)\} \tau \in \{0.1, 0.5, 0.9\}$$

- **Hierarchical pooled index :**

$$P_{z,t} = \text{coalesce}(P_{z, \text{traw}}, P_c(z), t_{\text{cty}}, P_s(z), t_{\text{st}})$$

Structural Design of Market Engine

It is a multi-dependant system interconnecting various modules which can be seen in the codebase.

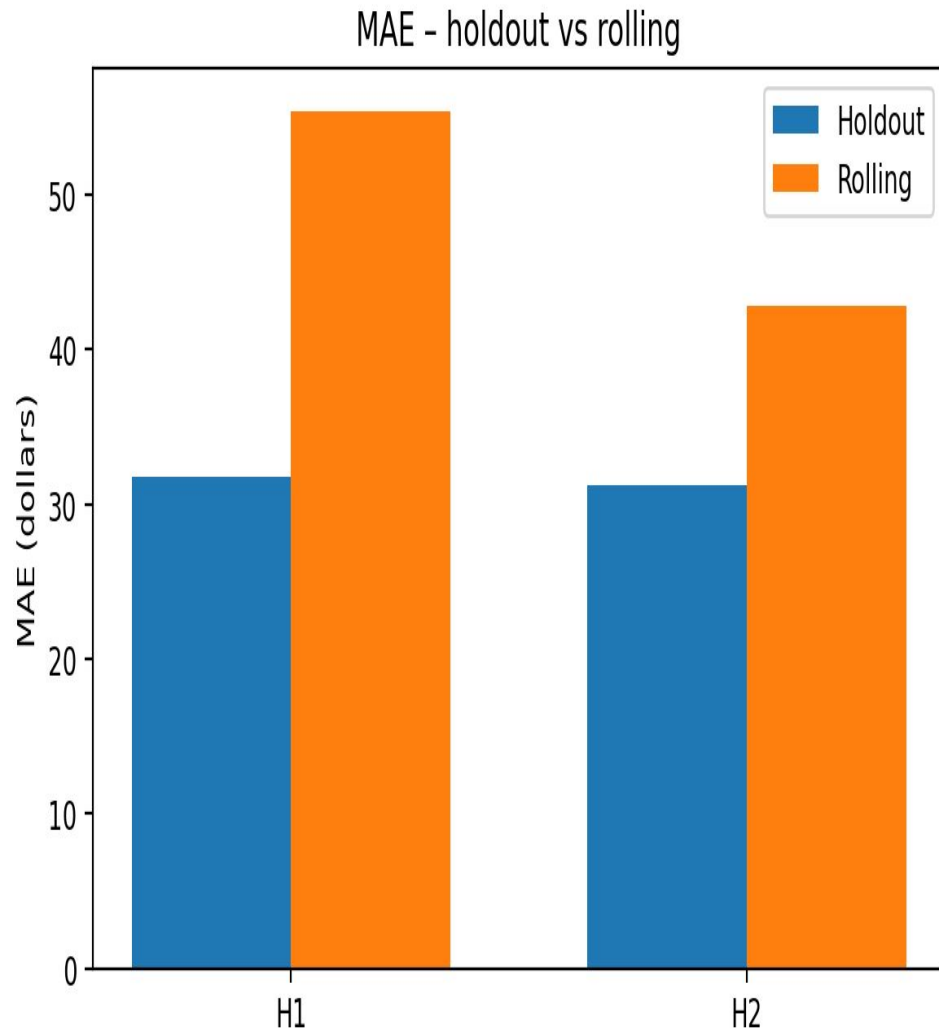


“Supporting information and evidence”

Low error on holdout : MAE is $\approx$ **\$31–32** for both 1-month (H1) and 2-month (H2) horizons.

Stress-tested over time: In rolling windows that span different market regimes, MAE stays in the **\$40–55** range, showing the model doesn’t break when the market shifts.

Actionable accuracy: Errors of $\sim$ \$30–40 at the ZIP $\times$ month level are small relative to median home prices, making these forecasts usable for pricing, budgeting, and risk planning.





Applying the Market Engine to a New Dataset: County & ZIP Results

Inference of our market engine on a separate builder dataset, with accuracy reported at county and ZIP level.

County	N	MAE H1	MAE H2
Wake County	29	102.88	71.65
Prince William County	26	150.59	105.85
Lake County	25	90.03	60.26
Aiken County	23	88.85	62.35
Prince Georges County	21	131.76	83.01
Brevard County	19	91.40	65.64
Spartanburg County	18	107.68	54.99
Polk County	15	82.93	59.61
Fairfax County	14	163.76	87.57
Columbia County	12	96.92	62.52

ZIP code	N	MAE H1	MAE H2
29801	16	85.15	67.35
30102	14	95.69	64.66
27587	13	89.07	72.48
32796	10	93.81	62.94
33851	10	81.34	61.75
22026	10	133.60	102.78
29909	9	124.56	78.66
30814	8	95.27	63.43
25438	8	106.31	62.24
20112	7	157.29	111.37

Property Engine: Latent Manifold Estimation

Decomposes each home's log price into:

- **Intrinsic value** from tabular attributes (beds, baths, sqft, year built, etc.)
- A smooth **neighborhood desirability field** over latitude/longitude.
- Mimics how an appraiser works: start with structure-driven value, then adjust up/down for the micro-location.
- Provides **house-specific premiums/discounts** relative to the local market and the ZIP forecast from the Market Engine.

Core Equation & Geometry

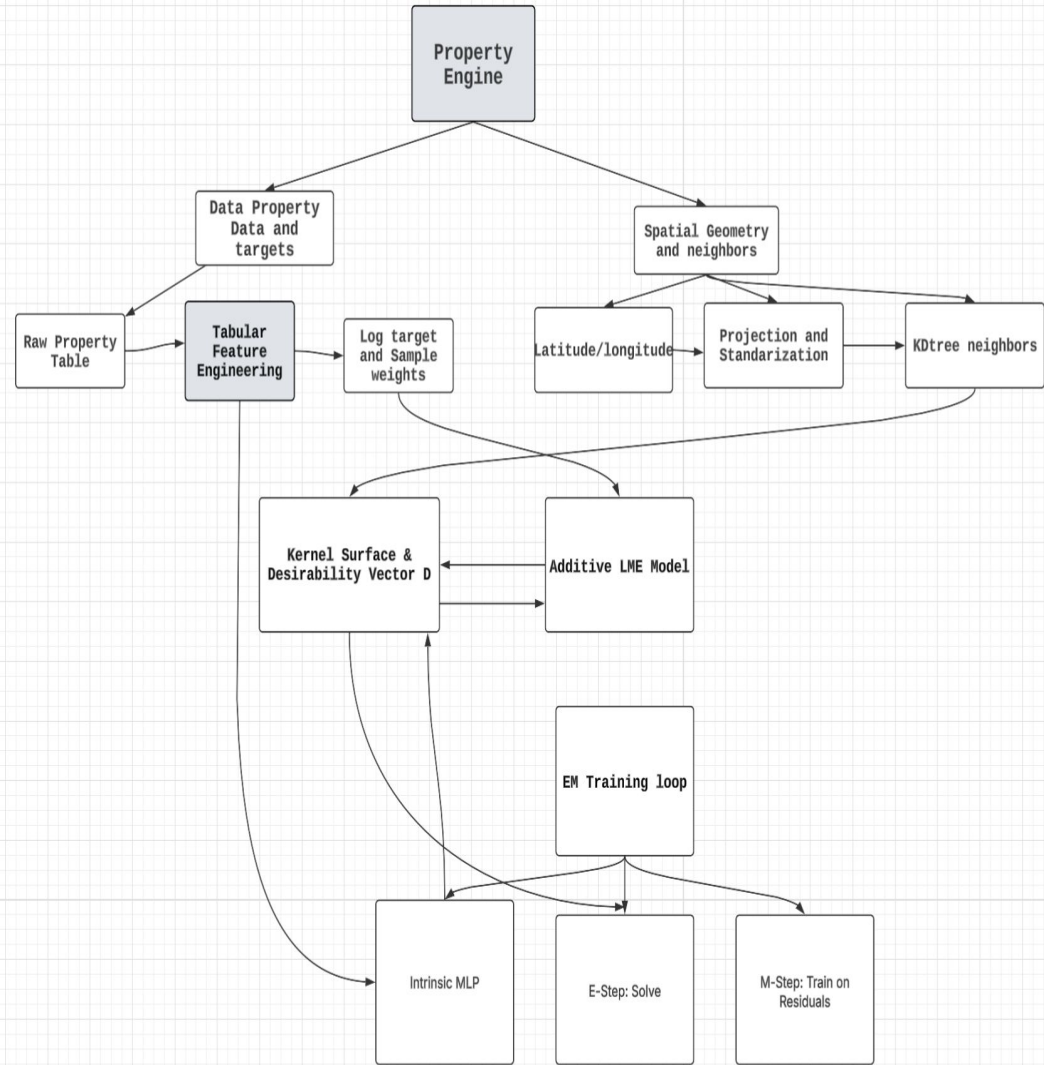
Modelling each home :

$$y_i = m_i + h_i + \varepsilon_i, \quad m_i = G(x_i^{(\text{param})}; W), \quad h_i = H(s_i; D),$$

$$x_i = R(\lambda_i - \bar{\lambda}) \cos \bar{\phi}, \quad \bar{y}_i = R(\phi_i - \bar{\phi}), \quad R = 6.371 \times 10^6 \text{ m.}$$

Structural Design of Property Engine

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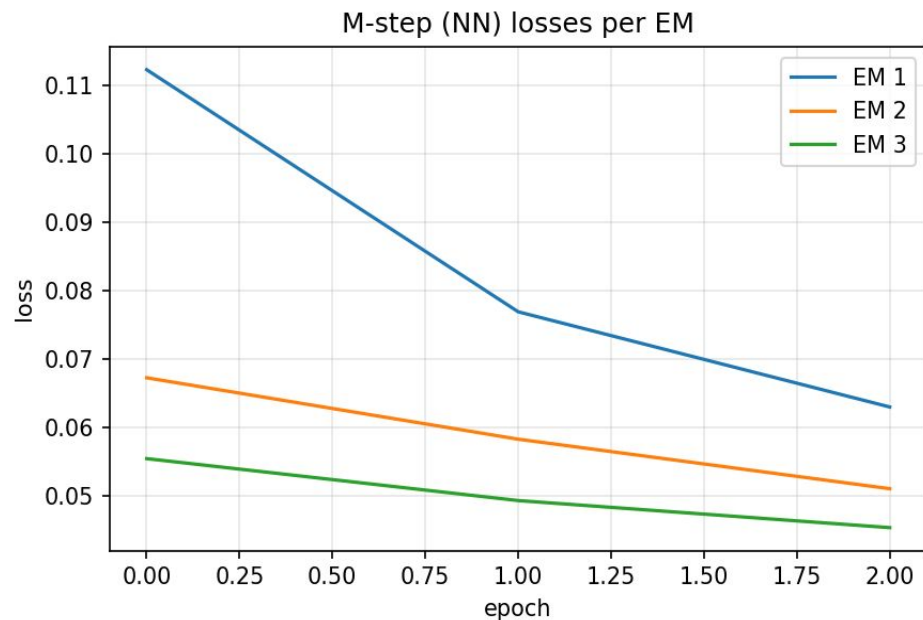


“Supporting information and evidence”

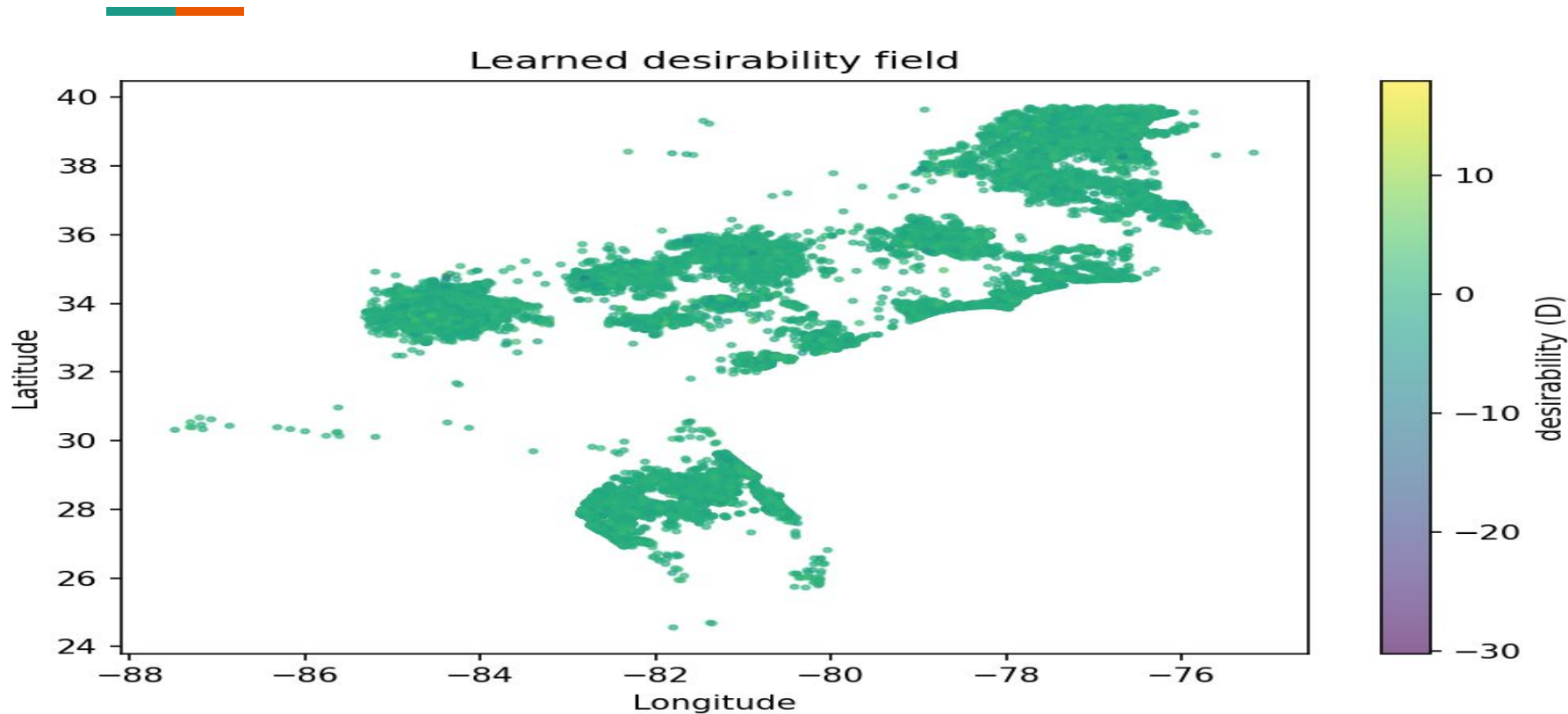
Current implementation accuracy: On our spatial test split,

- **18.1%** of homes are within **5%** price error,
 - **35.45%** within **10%**,
 - **50.62%** within **15%**,
- with **median AbsRel** ≈ 0.148 . The corresponding train metrics are 25.54%, 49.10%, 67.54% and median AbsRel 0.1022.

	< 5%	< 10%	< 15%	Median abs rel
Train	25.54%	49.10%	67.54%	0.1022
Test	18.10%	35.45%	50.62%	0.1479



Desirability learned by Property engine



End-to-end system:

Market layer: The Market Engine produces **probabilistic ZIP×Month indices** 1–2 months ahead, with downside/upside bands for each geography.

Property layer: The Property Engine takes these market indices as context and adds **home-specific intrinsic value and neighborhood desirability**, producing a full price distribution for each address.

Decision layer: On top of these outputs we can build tools for **land acquisition, release timing, list-price recommendations, and portfolio risk dashboards** (e.g., which ZIPs or projects are over-exposed).

Feedback loop: As new transactions come in, the system **re-trains both engines**, tightening forecasts and updating neighborhood premiums in near real time.

Business View

Together, the two engines become a reusable “Neural estimate” platform for the builder: one model family owns **market trajectory and risk**, the other owns **micro-location and product-level pricing**, and both plug directly into existing budgeting, underwriting, and sales workflows.

Questions?
